

Monte Carlo Simulation Uge 3

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1 Sampling Uniformly from Regions

This section generalizes the idea of sampling uniformly from a region in $\mathbb{R}^n$. The goal is to sample points uniformly from a given region $B \subseteq \mathbb{R}^n$. Suppose that we sample $X = (X_1, X_2, \dots, X_n)$ uniformly from a region B . Then the joint density of X is given by

$$f_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n) = \frac{1}{|B|}, \quad (1)$$

where $|B|$ is the volume (or area, in the case of $\mathbb{R}^2$) of the region B . The density is constant in the region B and zero outside of B . We can use this to find marginal densities of the individual components X_i . For example, the marginal density of X_1 is given by

$$f_{X_1}(x_1) = \int_{B_{x_1}} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{X_1 X_2 \dots X_n}(x_1, x_2, \dots, x_n) dx_2 \dots dx_n = \int_{B_{x_1}} \frac{1}{|B|} dx_2 \dots dx_n = \frac{|B_{x_1}|}{|B|}. \quad (2)$$

This means that the marginal density of X_1 is proportional to the volume of the region B_{x_1} , which is the slice of the region B at a fixed value of x_1 . This can be generalized to any component X_i .

1.1 Task 12 Acceptance-Rejection

We illustrate using Task 12 as an example. We are given the distribution function $f(x) = Cx^{-3/2} \frac{1}{\sqrt{2\pi}} e^{-(x-1)^2/2x}$, and we wish to find a dominating distribution $g(x)$ such that $f(x) \leq Mg(x)$ for some constant M . If we expand the expression for $f(x)$ and disregard the constant, as this can always be absorbed into the constant M , we have that

$$f(x) \propto x^{3/2} e^{-(x-1)^2/2x} = x^{-3/2} e^{-x/2} e^{1/2} e^{-1/2x} \propto x^{-3/2} e^{-x/2} e^{-1/2x}. \quad (1)$$

The behaviour near $x = 0$ can be assessed graphically, however the tail behaviour as $x \rightarrow \infty$ can be assessed analytically. We have that $x^{-3/2} \rightarrow 0$ as $x \rightarrow \infty$, and $e^{-1/2x} \rightarrow 1$ as $x \rightarrow \infty$. So we have to match the tail behaviour of $e^{-x/2}$. Thus if we choose the exponential distribution $g(x) = \lambda e^{-\lambda x}$, we have that $f(x) \leq Mg(x)$ for some constant M and $\lambda = 1/2$. Graphically, we can see that $M \approx 2.7$.

Let (X, Y) be uniformly distributed on the region $B = \{(x, y) \in \mathbb{R}^2 : 0 \leq y \leq f(x)\}$. Where $f(x)$ is a distribution function. We then wish to find the marginal density of X . We have that

$$|B| = \int \int_B \mathbb{1}(0 \leq y \leq f(x)) dy dx = \int_{-\infty}^{\infty} \int_0^{f(x)} dy dx = \int_{-\infty}^{\infty} f(x) dx = 1, \quad (2)$$

Now for a fixed x , we have that

$$f_X(x) = \frac{\int_0^{f(x)} f_{XY}(x, y) dy}{|B|} = \frac{\int_0^{f(x)} 1 dy}{1} = f(x). \quad (3)$$

This shows us that if we sample uniformly from the region below the curve of a distribution function $f(x)$, then the marginal density of X is given by $f(x)$. This is the basis for the acceptance-rejection method.

2 Transformation of Variables

Suppose that we have a vector of random variables $X = (X_1, X_2, \dots, X_n)$ with joint density $f_X(x)$. We wish to find the joint density of a new vector of random variables $Y = (Y_1, Y_2, \dots, Y_n)$, where $Y = g(X)$ for some function g . If g is a one-to-one transformation, then we can find the joint density of Y using the change of variables formula. Let $h = g^{-1}$ be the inverse transformation. Then the joint density of Y is given by

$$f_Y(y) = f_X(h(y)) |\det J_h(y)|, \quad y \in \mathbb{R}^n. \quad (1)$$

2.1 Task 13

We are given the transformation of (X, Y) to (V, U) , where $X = \sqrt{V}\cos(U)$ and $Y = \sqrt{V}\sin(U)$, Where $V > 0$ and $0 < U < \pi/2$. X and Y independent and standard normal. We wish to find the joint density of (V, U) . Note that the trasform implies that

$$X^2 + Y^2 = V$$

And since (X, Y) is jointly standard normal, we have that

$$f_{X,Y}(x, y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2+y^2)} = \frac{1}{2\pi} e^{-\frac{1}{2}V}$$

$$h(v, u) = (\sqrt{v}\cos(u), \sqrt{v}\sin(u)) = (x, y), \quad (1)$$

and the Jacobian matrix of h is

$$J_h(v, u) = \begin{pmatrix} \frac{\partial x}{\partial v} & \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial v} & \frac{\partial y}{\partial u} \end{pmatrix} = \begin{pmatrix} \frac{1}{2\sqrt{v}}\cos(u) & -\sqrt{v}\sin(u) \\ \frac{1}{2\sqrt{v}}\sin(u) & \sqrt{v}\cos(u) \end{pmatrix}. \quad (2)$$

The determinant of the Jacobian matrix is

$$\det J_h(v, u) = \frac{1}{2\sqrt{v}}\cos(u) \cdot \sqrt{v}\cos(u) - (-\sqrt{v}\sin(u)) \cdot \frac{1}{2\sqrt{v}}\sin(u) = \frac{1}{2}\cos^2(u) + \frac{1}{2}\sin^2(u) = \frac{1}{2}. \quad (3)$$

Therefore, the joint density of (V, U) is

$$f_{V,U}(v, u) = f_{X,Y}(h(v, u)) |\det J_h(v, u)| = \frac{1}{2\pi} e^{-\frac{1}{2}V} \cdot \frac{1}{2}. \quad (4)$$

We see that this is a product of two functions, one in v and one in u , which means that V and U are independent. More specifically, we have that V is exponentially distributed with rate parameter $\lambda = 1/2$, and U is uniformly distributed on the interval $(0, \pi/2)$. This then means that (V, U) are points at distance $\sqrt{V}$ from the origin, and the angle U is uniformly distributed on the interval $(0, \pi/2)$.

Suppose we want point uniformly from the unit sphere. Then we are interested in points (A, B, C) such that $A^2 + B^2 + C^2 = 1$. We have just seen above that independant standard normals are direction invariant, since the joint density functions becomes a function of the random variables through $\|(X, Y, Z)\|^2$. As such if we normalize the vector (X, Y, Z) , such that $(X, Y, Z) \rightarrow \frac{(X, Y, Z)}{\|(X, Y, Z)\|}$, then we have a point uniformly distributed on the unit sphere, where $A = \frac{X}{\|(X, Y, Z)\|}$, $B = \frac{Y}{\|(X, Y, Z)\|}$, and $C = \frac{Z}{\|(X, Y, Z)\|}$. If we substitute into the equation for the unit sphere, we have that

$$A^2 + B^2 + C^2 = \frac{X^2}{\|(X, Y, Z)\|^2} + \frac{Y^2}{\|(X, Y, Z)\|^2} + \frac{Z^2}{\|(X, Y, Z)\|^2} = \frac{X^2 + Y^2 + Z^2}{\|(X, Y, Z)\|^2} = 1. \quad (5)$$

3 Task 15

In the following we are interested in finding a bounding box for the region $B = \{(x_1, x_2) \in \mathbb{R}^2 : 0 \leq x_1 \leq \sqrt{g(x_1/x_2)}\}$, where $f(x)$ is a distribution function. We have a given candidate density function $g(x) = xe^{-x}$. The steps we need to follow are:

1. Find the supremum of $\sqrt{g(x)}$, this will give us the upper bound for x_1 .
2. Define $x = x_2/x_1$, note that $x \in \mathbb{R}$, can be both negative and positive.
3. Find the bounds for x_2 , since $x_2 = xx_1$, it follows from the inequality in B , and the fact that x can be both negative and positive, that $-\sup_{x \in \mathbb{R}} x\sqrt{g(x)} < x_2 < \sup_{x \in \mathbb{R}} x\sqrt{g(x)}$.

For the the function $g(x) = xe^{-x}$, we find the supremum of $\sqrt{g(x)}$ to be $\frac{1}{\sqrt{e}}$. For the bounds of x_2 , we have that $x_2 = xx_1$, and since x can be both negative and positive, we have that $-\sup_{x \in \mathbb{R}} x\sqrt{g(x)} < x_2 < \sup_{x \in \mathbb{R}} x\sqrt{g(x)}$. We find the supremum of $x\sqrt{g(x)}$ to be ≈ 1.16 . Thus we have that the bounding box is given by

$$A = \{(x_1, x_2) \in \mathbb{R}^2 : 0 \leq x_1 \leq \frac{1}{\sqrt{e}}, -1.16 < x_2 < 1.16\}. \quad (1)$$

Thus we need to sample from points $U \sim Unif(0, \frac{1}{\sqrt{e}})$ and $V \sim Unif(-1.16, 1.16)$, and accept the points that fall in the region B . Note for this particular function, negative values of x_2 will create point in the complex plane, so we can further restrict the bounding box to be $A = \{(x_1, x_2) \in \mathbb{R}^2 : 0 \leq x_1 \leq \frac{1}{\sqrt{e}}, 0 < x_2 < 1.16\}$.